Mahindra University

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Mahindra University Hyderabad

Ecole Centrale School of Engineering

Minor-I Examination

Program: B. Tech. Branch: CAM Year: II Semester: I

Subject: Design and Analysis of Algorithms (MA 2107)

Date: 12/09/2025 **Start Time: 2:00 PM**

Time Duration: 1.5 Hours **Max. Marks: 30**

Instructions:

1. There are 5 questions, all of which are compulsory.
2. Justify your response whenever necessary. Guesswork will not be considered in evaluation.
3. Answers without proper numbering will not be evaluated.

1. Answer the following questions: **8 M**

   a) State Master Theorem 3M

   b) Solve the following recurrence relation:

$$T(n) = 4T\left(\frac{n}{2}\right) + cn$$

5M

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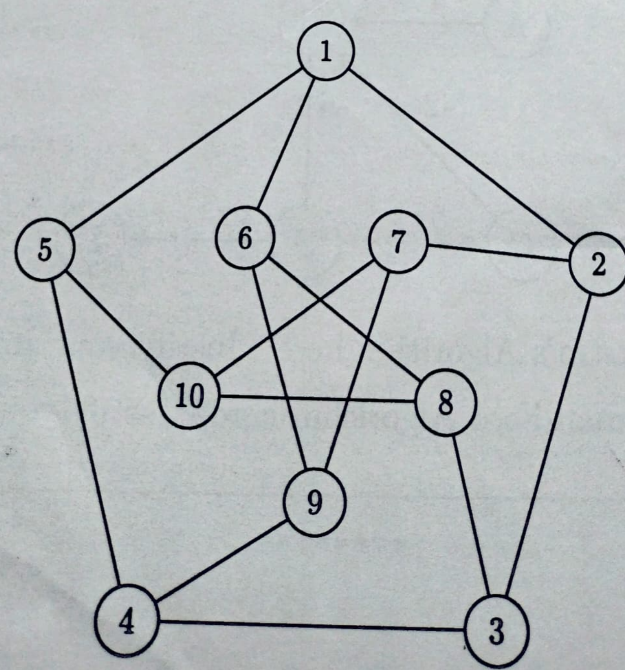
2. Draw the **BFS** and **DFS** tree of the following graph starting at vertex 5: **5 M**



3. Given the following list of numbers: **5 M**

$$42,\ 17,\ 8,\ 23,\ 56,\ 34$$

a. Arrange the list in ascending order using the **Selection Sort** algorithm. Show the steps. 3M

b. Perform a **Binary Search** on the sorted list to find the number 23. Show the steps. 2M

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4. Consider the following Directed Acyclic Graph (DAG) represented by its adjacency matrix: **8 M**

$$A = \begin{bmatrix} 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

The vertex set of the DAG is given by: $V = \{A, B, C, D, E, F\}$. Here, $A(i, j) = 1$ indicates a directed edge from vertex i to vertex j.

a. Draw the DAG. 2M

b. Write a **topological ordering** of the vertices. Is the topological ordering unique? Justify. 3M

c. Find the **longest path** starting from vertex A. 3M

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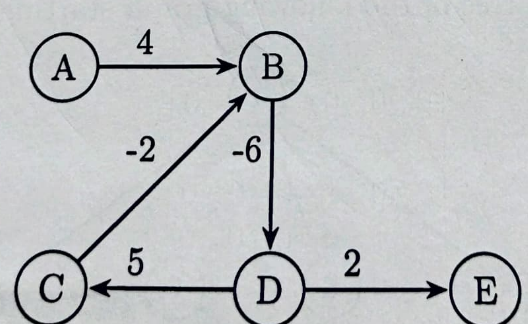
5. Consider the following graph: **4 M**



a. Can we apply Dijkstra's Algorithm here? Justify your answer. 2M

b. Can we apply Bellman-Ford Algorithm here? Justify your answer. 2M

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